

Concept 5 | Inequalities

English Student Edition • Focused formulas and guided practice

Formula opener

Square rule

$$(x - y)^2 \geq 0$$

Order

$$0 < a < b \Rightarrow a^2 < b^2$$

AM-GM

$$\frac{a+b}{2} \geq \sqrt{ab}$$

Classified Practice

Inequalities

Q001 **Written**

Under the condition $a > b$, $x > y$, prove that $2(ax + by) > (a + b)(x + y)$.

Q002 MCQ**If $x > y$, prove that $3x + y > 2(x + y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q003 MCQ

If $x > y$, prove that $3x + 2y > 2x + 3y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $x - y$

D $-x + y$

Q004 MCQ**If $x > y$, prove that $3(x + y) > 2(x + 2y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Classified Practice

Inequalities

Q005 MCQ

If $x > y$, prove that $3x + 4y > 2x + 5y$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y + 1$

B $x - y - 1$

C $-x + y$

D $x - y$

Q006 MCQ**If $x > y$, prove that $3x + 5y > 2(x + 3y)$.****If $x > y$, the positive difference between the two sides is:****Choose the correct option.**

A $x - y + 1$

B $x - y$

C $x - y - 1$

D $-x + y$

Classified Practice

Inequalities

Q007 Written

Under the condition $a > b$, $c > d$, prove that $ac + bd > ad + bc$.

Q008 Written

Under the condition $a > c$, $b > d$, prove that $ab + cd > ad + bc$.

Classified Practice

Inequalities

Q009 MCQ

If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$.

If $a > b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - bc + bd + 1$

B $ac - ad - bc + bd - 1$

C $-(a - b)(c - d)$

D $(a - b)(c - d)$

Q010 MCQ

If $a > 2b$ and $c > d$, prove that $ac + 2bd > ad + 2bc$.

If $a > 2b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $ac - ad - 2bc + 2bd + 1$

B $ac - ad - 2bc + 2bd - 1$

C $(a - 2b)(c - d)$

D $-(a - 2b)(c - d)$

Classified Practice

Inequalities

Q011 MCQ

If $a > 3b$ and $c > d$, prove that $ac + 3bd > ad + 3bc$.

If $a > 3b$ and $c > d$, the positive difference between the two sides is:

Choose the correct option.

A $(a - 3b)(c - d)$

B $ac - ad - 3bc + 3bd + 1$

C $ac - ad - 3bc + 3bd - 1$

D $-(a - 3b)(c - d)$

Q012 Written

Under the condition $x > y$, prove that $3x - 4y > 2x - 3y$.

Classified Practice

Inequalities

Q013 Written

Under the condition $x > 3$, $y > -5$, prove that $xy + 5x > 3y + 15$.

Q014 Written

Under the condition $x > 2$, $y > 1$, prove that $xy + 2 > x + 2y$.

Classified Practice

Inequalities

Q015 Written

Under the condition $a > 1$, $b > 1$, prove that $ab + 1 > a + b$.

Q016 Written

Under the condition $a > b > x > y$, prove that $ax + by > ay + bx$.

Classified Practice

Inequalities

Q017 Written

Under the condition $a > b > c > 0$, $x > y > z > 0$, prove that $ax + by + cz > ay + bz + cx$.

Classified Practice

Inequalities

Q018 Challenge

If $a > b$ and $x > y$, prove that $2(ax + by) > 2(ay + bx)$.

If $a > b$ and $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $2ax - 2ay - 2bx + 2by + 1$

B $2ax - 2ay - 2bx + 2by - 1$

C $-2(a - b)(x - y)$

D $2(a - b)(x - y)$

Classified Practice

Inequalities

Q019 Written

Prove that $a^2 + 4b^2 \geq 4ab$.

Q020 MCQ

Under a,b real, prove that $a^2 + 4b^2 \geq 4ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 4ab + 4b^2 + 1$

B $(a - 2b - 1)(a - 2b + 1)$

C $-(a - 2b)^2$

D $(a - 2b)^2$

Classified Practice

Inequalities

Q021 MCQ

Under a, b real, prove that $a^2 + 9b^2 \geq 6ab$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $a^2 - 6ab + 9b^2 + 1$
B $(a - 3b - 1)(a - 3b + 1)$
C $(a - 3b)^2$
D $-(a - 3b)^2$

Q022 MCQ

Under a,b real, prove that $a^2 + 16b^2 \geq 8ab$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $a^2 - 8ab + 16b^2 + 1$

B $(a - 4b)^2$

C $(a - 4b - 1)(a - 4b + 1)$

D $-(a - 4b)^2$

Classified Practice

Inequalities

Q023 MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Q024 MCQ

Under a,b real, prove that $a^2 + 36b^2 \geq 12ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 12ab + 36b^2 + 1$

B $(a - 6b - 1)(a - 6b + 1)$

C $(a - 6b)^2$

D $-(a - 6b)^2$

Classified Practice

Inequalities

Q025 MCQ

Under a, b real, prove that $a^2 + 49b^2 \geq 14ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $a^2 - 14ab + 49b^2 + 1$

B $(a - 7b)^2$

C $(a - 7b - 1)(a - 7b + 1)$

D $-(a - 7b)^2$

Q026 Written**Prove that** $a^2 + 5b^2 \geq 4ab$.

Classified Practice

Inequalities

Q027 Written

Prove that $9x^2 \geq y(6x - y)$.

Q028 Written**Prove that** $2(x^2 + y^2) \geq (x + y)^2$.

Classified Practice

Inequalities

Q029 Written

Prove that $(2x - y)^2 \geq 3(x^2 - y^2)$.

Q030 Written**Prove that** $(2a + 3b)^2 \geq 12ab$.

Classified Practice

Inequalities

Q031 MCQ

Under x, y real, prove that $4x^2 \geq y(4x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(2x - y)^2$

B $4x^2 - 4xy + y^2 + 1$

C $(2x - y - 1)(2x - y + 1)$

D $-(2x - y)^2$

Q032 MCQ

Under x, y real, prove that $9x^2 \geq y(6x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $9x^2 - 6xy + y^2 + 1$
- B $(3x - y - 1)(3x - y + 1)$
- C $(3x - y)^2$
- D $-(3x - y)^2$

Classified Practice

Inequalities

Q033 MCQ

Under x, y real, prove that $16x^2 \geq y(8x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $16x^2 - 8xy + y^2 + 1$

B $(4x - y - 1)(4x - y + 1)$

C $-(4x - y)^2$

D $(4x - y)^2$

Q034 MCQ

Under x, y real, prove that $25x^2 \geq y(10x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $25x^2 - 10xy + y^2 + 1$

B $(5x - y - 1)(5x - y + 1)$

C $-(5x - y)^2$

D $(5x - y)^2$

Classified Practice

Inequalities

Q035 MCQ

Under x, y real, prove that $36x^2 \geq y(12x - y)$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $(6x - y)^2$

B $36x^2 - 12xy + y^2 + 1$

C $(6x - y - 1)(6x - y + 1)$

D $-(6x - y)^2$

Q036 MCQ**Under x, y real, prove that $x^2 + y^2 \geq 2xy$ select the correct factorisation of LHS - RHS:****Choose the correct option.**

A $x^2 - 2xy + y^2 + 1$

B $(x - y - 1)(x - y + 1)$

C $(x - y)^2$

D $-(x - y)^2$

Classified Practice

Inequalities

Q037 **Written****Prove that** $x^2 + y^2 \geq 2(x - y - 1)$.

Q038 Written**Prove that** $x^2 + 2 > 2x$.

Classified Practice

Inequalities

Q039 Written

Prove that $x^2 + y^2 - 6x + 8y + 25 \geq 0$.

Q040 Written**Prove that** $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q041 **Written****Prove that** $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$.

Classified Practice

Inequalities

Q042 **Challenge**

For real x, y , prove that $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$.

Classified Practice

Inequalities

Q043 Written

Prove that $\sqrt{\frac{a^2+b^2}{2}} \geq \frac{a+b}{2}$.

Classified Practice

Inequalities

Q044 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{b^2}{2}} \geq \frac{a}{2} + \frac{b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a-b)^2}{4}$

B $\frac{a^2 - 2ab + b^2 + 4}{4}$

C $\frac{(a-b-2)(a-b+2)}{4}$

D $-\frac{(a-b)^2}{4}$

Classified Practice

Inequalities

Q045 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 2b^2} \geq \frac{a}{2} + b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 2b)^2}{4}$

B $\frac{a^2 - 4ab + 4b^2 + 4}{4}$

C $\frac{(a - 2b - 2)(a - 2b + 2)}{4}$

D $-\frac{(a - 2b)^2}{4}$

Classified Practice

Inequalities

Q046 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{9b^2}{2}} \geq \frac{a}{2} + \frac{3b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{a^2 - 6ab + 9b^2 + 4}{4}$

B $\frac{(a - 3b - 2)(a - 3b + 2)}{4}$

C $\frac{(a - 3b)^2}{4}$

D $-\frac{(a - 3b)^2}{4}$

Classified Practice

Inequalities

Q047 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 8b^2} \geq \frac{a}{2} + 2b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 4b)^2}{4}$

B $\frac{a^2 - 8ab + 16b^2 + 4}{4}$

C $\frac{(a - 4b - 2)(a - 4b + 2)}{4}$

D $-\frac{(a - 4b)^2}{4}$

Classified Practice

Inequalities

Q048 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{25b^2}{2}} \geq \frac{a}{2} + \frac{5b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 5b)^2}{4}$
- B $\frac{a^2 - 10ab + 25b^2 + 4}{4}$
- C $\frac{(a - 5b - 2)(a - 5b + 2)}{4}$
- D $-\frac{(a - 5b)^2}{4}$

Classified Practice

Inequalities

Q049 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 18b^2} \geq \frac{a}{2} + 3b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{a^2 - 12ab + 36b^2 + 4}{4}$

B $\frac{(a - 6b)^2}{4}$

C $\frac{(a - 6b - 2)(a - 6b + 2)}{4}$

D $-\frac{(a - 6b)^2}{4}$

Classified Practice

Inequalities

Q050 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 7b)^2}{4}$
- B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$
- C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$
- D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Q051 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + 32b^2} \geq \frac{a}{2} + 4b$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $\frac{(a - 8b)^2}{4}$

B $\frac{a^2 - 16ab + 64b^2 + 4}{4}$

C $\frac{(a - 8b - 2)(a - 8b + 2)}{4}$

D $-\frac{(a - 8b)^2}{4}$

Classified Practice

Inequalities

Q052 Written

Prove that $\sqrt{2(a+b)} \geq \sqrt{a} + \sqrt{b}$.

Q053 Written**Prove that** $a + 2 \geq \sqrt{3 + 6a}$.

Classified Practice

Inequalities

Q054 Written

Prove that $\sqrt{2(a^2 + b^2)} \geq a + b$.

Q055 Written

Prove that $\sqrt{\frac{a+b}{2}} \geq \frac{\sqrt{a}+\sqrt{b}}{2}$.

Classified Practice

Inequalities

Q056 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $2\sqrt{a}\sqrt{b} + 1$

B $2\sqrt{a}\sqrt{b}$

C $2\sqrt{a}\sqrt{b} - 1$

D $-2\sqrt{a}\sqrt{b}$

Q057 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 2\sqrt{b} > \sqrt{a + 4b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $4\sqrt{a}\sqrt{b}$

B $4\sqrt{a}\sqrt{b} + 1$

C $4\sqrt{a}\sqrt{b} - 1$

D $-4\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q058 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 3\sqrt{b} > \sqrt{a + 9b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $6\sqrt{a}\sqrt{b} + 1$

B $6\sqrt{a}\sqrt{b} - 1$

C $6\sqrt{a}\sqrt{b}$

D $-6\sqrt{a}\sqrt{b}$

Q059 MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

A $8\sqrt{a}\sqrt{b} + 1$

B $8\sqrt{a}\sqrt{b}$

C $8\sqrt{a}\sqrt{b} - 1$

D $-8\sqrt{a}\sqrt{b}$

Classified Practice

Inequalities

Q060 Written

Prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Q061 Written**Prove that** $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$.

Classified Practice

Inequalities

Q062 Written

Prove that $\sqrt{a} + \sqrt{b} > \sqrt{a+b}$.

Q063 Written**Prove that** $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$.

Classified Practice

Inequalities

Q064 Written

Prove that $\sqrt{a-b} > \sqrt{a} - \sqrt{b}$.

Q065 Written

If $a > b > 0$, prove that $\sqrt{a-b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$.

Classified Practice

Inequalities

Q066 Challenge

For $a, b > 0$, prove that $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$.

Classified Practice

Inequalities

Q067 Written

Prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Q068 Written**Prove that** $\frac{b}{3a} + \frac{12a}{b} \geq 4$.

Classified Practice

Inequalities

Q069 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{a^2 + b^2}{ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(a - b)^2}{ab}$

B $\frac{a^2 - ab + b^2}{ab}$

C $\frac{a^2 - 3ab + b^2}{ab}$

D $-\frac{(a - b)^2}{ab}$

Q070 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{4a^2 + b^2}{2ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{4a^2 - 2ab + b^2}{2ab}$

B $\frac{4a^2 - 6ab + b^2}{2ab}$

C $\frac{(2a - b)^2}{2ab}$

D $-\frac{(2a - b)^2}{2ab}$

Classified Practice

Inequalities

Q071 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{9a^2 + b^2}{3ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{9a^2 - 3ab + b^2}{3ab}$

B $\frac{(3a - b)^2}{3ab}$

C $\frac{9a^2 - 9ab + b^2}{3ab}$

D $-\frac{(3a - b)^2}{3ab}$

Q072 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{16a^2 + b^2}{4ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{(4a - b)^2}{4ab}$

B $\frac{16a^2 - 4ab + b^2}{4ab}$

C $\frac{16a^2 - 12ab + b^2}{4ab}$

D $-\frac{(4a - b)^2}{4ab}$

Classified Practice

Inequalities

Q073 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{25a^2 + b^2}{5ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{25a^2 - 5ab + b^2}{5ab}$
- B $\frac{25a^2 - 15ab + b^2}{5ab}$
- C $-\frac{(5a - b)^2}{5ab}$
- D $\frac{(5a - b)^2}{5ab}$

Q074 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{36a^2 + b^2}{6ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{36a^2 - 6ab + b^2}{6ab}$

B $\frac{36a^2 - 18ab + b^2}{6ab}$

C $-\frac{(6a - b)^2}{6ab}$

D $\frac{(6a - b)^2}{6ab}$

Classified Practice

Inequalities

Q075 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{49a^2 + b^2}{7ab} \geq 2$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{49a^2 - 7ab + b^2}{7ab}$
- B $\frac{49a^2 - 21ab + b^2}{7ab}$
- C $\frac{(7a - b)^2}{7ab}$
- D $-\frac{(7a - b)^2}{7ab}$

Q076 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{64a^2 + b^2}{8ab} \geq 2$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{64a^2 - 8ab + b^2}{8ab}$

B $\frac{64a^2 - 24ab + b^2}{8ab}$

C $-\frac{(8a - b)^2}{8ab}$

D $\frac{(8a - b)^2}{8ab}$

Classified Practice

Inequalities

Q077 Written

Prove that $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$.

Q078 Written**Prove that** $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$.

Classified Practice

Inequalities

Q079 Written

Prove that $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$.

Q080 Written**Prove that** $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$.

Classified Practice

Inequalities

Q081 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+b)^2}{ab} \geq 4$ select the correct factorisation of LHS - RHS:

Choose the correct option.

- A $\frac{a^2 - ab + b^2}{ab}$
- B $\frac{a^2 - 3ab + b^2}{ab}$
- C $-\frac{(a-b)^2}{ab}$
- D $\frac{(a-b)^2}{ab}$

Q082 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Classified Practice

Inequalities

Q083 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+9b)(9a+b)}{ab} \geq 100$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{9a^2 - 17ab + 9b^2}{ab}$

B $\frac{9a^2 - 19ab + 9b^2}{ab}$

C $\frac{9(a-b)^2}{ab}$

D $-\frac{9(a-b)^2}{ab}$

Q084 MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+16b)(16a+b)}{ab} \geq 289$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $\frac{16a^2 - 31ab + 16b^2}{ab}$

B $\frac{16a^2 - 33ab + 16b^2}{ab}$

C $-\frac{16(a-b)^2}{ab}$

D $\frac{16(a-b)^2}{ab}$

Classified Practice

Inequalities

Q085 Written

Prove that $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$.

Q086 Written**Prove that $a + \frac{4}{a} \geq 4$.**

Classified Practice

Inequalities

Q087 Written

Prove that $3ab + \frac{4}{ab} \geq 4\sqrt{3}$.

Q088 Written**Prove that** $a + b + \frac{9}{a+b} \geq 6$.

Classified Practice

Inequalities

Q089 Written

A rectangle has perimeter 200 cm. Prove that its area $LW \leq 2500 \text{ cm}^2$.

Classified Practice

Inequalities

Q090 **Challenge**

For $a, b > 0$, prove that $\frac{b}{a} + \frac{a}{b} \geq 2$.

Classified Practice

Inequalities

Quiz MCQ 1 **Concept Quiz · MCQ**

If $x > y$, prove that $3x + y > 2(x + y)$.

If $x > y$, the positive difference between the two sides is:

Choose the correct option.

A $x - y$

B $x - y + 1$

C $x - y - 1$

D $-x + y$

Quiz MCQ 2 Concept Quiz · MCQ

Under a, b real, prove that $a^2 + 25b^2 \geq 10ab$ select the correct factorisation of LHS – RHS:

Choose the correct option.

A $(a - 5b)^2$

B $a^2 - 10ab + 25b^2 + 1$

C $(a - 5b - 1)(a - 5b + 1)$

D $-(a - 5b)^2$

Classified Practice

Inequalities

Quiz MCQ 3 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\sqrt{\frac{a^2}{2} + \frac{49b^2}{2}} \geq \frac{a}{2} + \frac{7b}{2}$.

After squaring the positive sides, the value of $\text{LHS}^2 - \text{RHS}^2$ is:

Choose the correct option.

- A $\frac{(a - 7b)^2}{4}$
- B $\frac{a^2 - 14ab + 49b^2 + 4}{4}$
- C $\frac{(a - 7b - 2)(a - 7b + 2)}{4}$
- D $-\frac{(a - 7b)^2}{4}$

Classified Practice

Inequalities

Quiz MCQ 4 Concept Quiz · MCQ

Under $a > 0$ and $b > 0$, prove that $\frac{(a+4b)(4a+b)}{ab} \geq 25$ select the correct factorisation of LHS - RHS:

Choose the correct option.

A $\frac{4a^2 - 7ab + 4b^2}{ab}$

B $\frac{4(a-b)^2}{ab}$

C $\frac{4a^2 - 9ab + 4b^2}{ab}$

D $-\frac{4(a-b)^2}{ab}$

Quiz WRITTEN 5 **Concept Quiz · Written**

If $x > y$, prove that $3x + 4y > 2x + 5y$.

Classified Practice

Inequalities

Quiz WRITTEN 6 Concept Quiz · Written**Under x, y real, prove that $x^2 + y^2 \geq 2xy$.**

Quiz WRITTEN 7 Concept Quiz · Written

Under $a > 0$ and $b > 0$, prove that $\sqrt{a} + 6\sqrt{b} > \sqrt{a + 36b}$.

Classified Practice

Inequalities

Quiz WRITTEN 8 Concept Quiz · Written

Under $t > 0$, prove that $\frac{t^2+1}{t} \geq 2$.

Answer key

Use the question number shown on each page.

Q001 $2(ax + by) > (a + b)(x + y)$

Q002 A

Q003 C

Q004 D

Q005 D

Q006 B

Q007 $ac + bd > ad + bc$

Q008 $ab + cd > ad + bc$

Q009 D

Q010 C

Q011 A

Q012 $3x - 4y > 2x - 3y$

Q013 $xy + 5x > 3y + 15$

Q014 $xy + 2 > x + 2y$

Q015 $ab + 1 > a + b$

Q016 $ax + by > ay + bx$

Q017 $ax + by + cz > ay + bz + cx$

Q018 D

Q019 $a^2 + 4b^2 \geq 4ab$

Q020 D

Q021 C

Q022 B

Q023 A

Q024 C

Q025 B

Q026 $a^2 + 5b^2 \geq 4ab$

Q027 $9x^2 \geq y(6x - y)$

Q028 $2(x^2 + y^2) \geq (x + y)^2$

Q029 $(2x - y)^2 \geq 3(x^2 - y^2)$

Q030 $(2a + 3b)^2 \geq 12ab$

Q031 A

Q032 C

Q033 D

Q034 D

Q035 A

Q036 C

Q037 $x^2 + y^2 \geq 2(x - y - 1)$

Q038 $x^2 + 2 > 2x$

Q039 $x^2 + y^2 - 6x + 8y + 25 \geq 0$

Q040 $x^2 + 2y^2 - 10x + 12y + 43 \geq 0$

Q041 $(a^2 + b^2)(x^2 + y^2) \geq (ax + by)^2$

Q042 0

Q043 $\sqrt{\frac{a^2 + b^2}{2}} \geq \frac{a + b}{2}$

Q044 A

Q045 A

Q046 C

Q047 A

Q048 A

Q049 B

Q050 A

Q051 A

Q052 $\sqrt{2(a + b)} \geq \sqrt{a} + \sqrt{b}$

Q053 $a + 2 \geq \sqrt{3 + 6a}$

Q054 $\sqrt{2(a^2 + b^2)} \geq a + b$

Q055 $\sqrt{\frac{a + b}{2}} \geq \frac{\sqrt{a} + \sqrt{b}}{2}$

Q056 B

Q057 A

Q058 C

Q059 B

Q060 $\sqrt{a} + 4\sqrt{b} > \sqrt{a + 16b}$

Q061 $2\sqrt{a} + \sqrt{b} > \sqrt{4a + b}$

Q062 $\sqrt{a} + \sqrt{b} > \sqrt{a + b}$

Q063 $2\sqrt{a} + 3\sqrt{b} > \sqrt{4a + 9b}$

Q064 $\sqrt{a - b} > \sqrt{a} - \sqrt{b}$

Q065 $\sqrt{a - b} < \sqrt{a} - \frac{b}{2\sqrt{a}}$

Q066 $\sqrt{a + 16b}$

Q067 $\frac{b}{a} + \frac{a}{b} \geq 2$

Q068 $\frac{b}{3a} + \frac{12a}{b} \geq 4$

Q069 A

Q070 C

Q071 B

Q072 A

Q073 D

Q074 D

Q075 C

Q076 D

Q077 $(a + b)\left(\frac{1}{a} + \frac{1}{b}\right) \geq 4$

Q078 $\left(x + \frac{1}{x}\right)\left(x + \frac{4}{x}\right) \geq 9$

Q079 $(a + 8b)\left(\frac{2}{a} + \frac{1}{b}\right) \geq 18$

Q080 $\left(a + \frac{3}{b}\right)\left(b + \frac{3}{a}\right) \geq 12$

Q081 D

Q082 B

Q083 C

Q084 D

Q085 $\left(a + \frac{1}{b}\right)\left(b + \frac{9}{a}\right) \geq 16$

Q086 $a + \frac{4}{a} \geq 4$

Q087 $3ab + \frac{4}{ab} \geq 4\sqrt{3}$

Q088 $a + b + \frac{9}{a + b} \geq 6$

Q089 $LW \leq 2500$

Q090 2

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 B

Quiz WRITTEN 5 $2x + 5y$

Quiz WRITTEN 6 $2xy$

Quiz WRITTEN 7 $\sqrt{a + 36b}$

Quiz WRITTEN 8 2